

Daily Combinatorics Drill #3 — Answers & Investigations

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Blocks & Gaps · Circular Arrangements · Repeated Letters · Forced-Order Tricks.

Section A Rapid Recognition

(≤15 s each)

1. In how many ways can 5 people stand in a row?

Answer: $5! = 120$

Method: Distinct objects in a row: factorial. Instant.

Investigate further: Sheet 1's opening question returns as pure recall — that is the point of Section A. If this took more than three seconds, redo sheet 1's A section tomorrow.

2. How many arrangements of the letters of LEMON?

Answer: $5! = 120$

Method: Five *distinct* letters — same question as A1 in costume.

Investigate further: Spot the isomorphism reflex: people in rows, distinct letters, distinct books are the same problem. The next question breaks the spell — watch what changes.

3. How many arrangements of the letters of BANANA?

Answer: $\frac{6!}{3!2!} = 60$

Method: Six letters: A×3, N×2, B×1. Divide 6! by the repeats: $\frac{720}{6 \cdot 2}$.

Investigate further: Why division works: the 3 A's can be permuted in 3! ways without changing the word, so each distinct word was counted exactly 3!2! times among the 6!. This “quotient out the symmetry” idea is the same one that will give circular arrangements in A4.

4. Four people around a round table (rotations the same)?

Answer: $(4 - 1)! = 6$

Method: Fix one person to kill the rotational symmetry; arrange the remaining 3 freely.

Investigate further: Alternative view: $\frac{4!}{4}$ — each circular arrangement is counted 4 times among row arrangements, once per rotation. Check both give 6. When would you also divide by 2? (When the table can be “flipped” — necklaces, handshake circles.)

5. Six in a row, Ali first and Zara last?

Answer: $4! = 24$

Method: Two positions forced, four people free.

Investigate further: Forced positions contribute factor 1 — cousins of sheet 2's forced aces. Variant: Ali *or* Zara first (and no other constraint): $2 \cdot 5! = 240$. Why can you not just double 24?

6. Three identical red flags and two identical blue flags in a row?

Answer: $\binom{5}{2} = 10$

Method: Only the *positions* of the blue flags matter: choose 2 places from 5. Equivalently $\frac{5!}{3!2!}$.

Investigate further: Repeated-letter arrangements and binomial coefficients are the same formula meeting in the middle: $\frac{5!}{3!2!} = \binom{5}{2}$. Every binary string question (sheet 2, B6) is a flag question. What is the three-colour version? (The multinomial coefficient — sheet 4.)

7. Arrangements of NOON?

Answer: $\frac{4!}{2!2!} = 6$

Method: Four letters, two pairs: $\frac{24}{4}$.

Investigate further: List all six to feel the formula: NOON, NONO, NNOO, OONN, ONON, ONNO. Any word whose repeat signature is (2, 2) has 6 arrangements — signature, not spelling, decides the count.

8. Six people around a round table (rotations the same)?

Answer: $(6 - 1)! = 120$

Method: Fix one person, arrange 5.

Investigate further: A dinner-party host claims seating 6 guests differently every night is possible for four months. Check: 120 nights $\approx$ four months. For how long could 8 guests last? (Nearly 14 years.)

9. True or false: 8 people round a table = 7 people in a row?

Answer: True — both are $7! = 5040$

Method: Circular $n = (n - 1)!$ = row arrangements of $n - 1$. Fixing one diner *is* reducing to a row of the rest.

Investigate further: TMUA-style equivalence spotting. Decide true/false: “ $2n$ people in a circle = n people in a row, for some $n > 1$.” ($(2n - 1)! = n!$ forces $2n - 1 = n$, impossible for $n > 1$ — false.)

10. Arrangements of TATTY?

Answer: $\frac{5!}{3!} = 20$

Method: Five letters with T $\times$ 3: divide by 3!.

Investigate further: Rank the words BANANA (60), TATTY (20), LEVEL (30), LEMON (120) by arrangement count *before* computing, using only the repeat structure. Fluency at reading the “signature” $(3, 2, 1)$ vs $(3, 1, 1)$ vs $(2, 2, 1)$ is the skill.

Section B Manipulation Drills

(≤ 50 s each)

1. Six in a row, Priya and Quinn together?

Answer: $2 \cdot 5! = 240$

Method: Glue the pair (block has 2 internal orders), arrange 5 objects.

Investigate further: Sheet 1 met this in Section C; now it is a B-level reflex — the spiral is deliberate. Speed target: write “ $2 \cdot 5!$ ” within five seconds of finishing reading.

2. Six in a row, Priya and Quinn *not* together?

Answer: $720 - 240 = 480$

Method: Complement of B1.

Investigate further: Reprove via gaps: seat the other 4 ($4!$), then place P and Q in different gaps of the 5 available: $4! \cdot 5 \cdot 4 = 480$. The gap method scales to “no two of *three* people adjacent” where complements get messy — see D5.

3. Arrangements of MISSISSIPPI?

Answer: $\frac{11!}{4!4!2!} = 34\,650$

Method: 11 letters: $I \times 4$, $S \times 4$, $P \times 2$, $M \times 1$. One division, no listing.

Investigate further: Compute it calculator-free the clean way: $\frac{11!}{4!4!2!} = \binom{11}{4} \binom{7}{4} \binom{3}{2} = 330 \cdot 35 \cdot 3$. Choosing positions letter-by-letter both proves the formula and organises the arithmetic.

4. Five boys, three girls in a row, no two girls adjacent?

Answer: $5! \cdot (6 \cdot 5 \cdot 4) = 120 \cdot 120 = 14\,400$

Method: Gap method: arrange the boys ($5!$), creating 6 gaps (including the two ends); place the three distinct girls in three *different* gaps, in order: $6 \cdot 5 \cdot 4$.

Investigate further: Why does gluing/complement fail here? “No two of three adjacent” has too many overlapping bad events. The gap method sidesteps inclusion–exclusion entirely — but inclusion–exclusion (sheet 5) will handle what gaps cannot.

5. Four men and four women in a row, alternating?

Answer: $2 \cdot 4! \cdot 4! = 1\,152$

Method: Two gender patterns (MWMW... or WMWM...), then the men fill their 4 slots in $4!$ ways and the women theirs in $4!$.

Investigate further: With 5 men and 4 women only one pattern survives (MWMWMWMWM): $5! \cdot 4! = 2880$, no factor 2. The factor-of-2 question is always: does the *pattern* have a choice, or is it forced?

6. Six around a round table, two particular people together?

Answer: $2 \cdot (5 - 1)! = 48$

Method: Glue the pair into a block: 5 objects round a table = $(5 - 1)!$, times 2 internal orders.

Investigate further: Row version was $2 \cdot 5! = 240$; circle version is $2 \cdot 4! = 48$. The ratio is 5 — exactly the number of rotations quotiented out. Check this ratio logic predicts the answer for 8 people with a glued pair.

7. Six-digit strings from 1, 1, 2, 2, 3, 3?

Answer: $\frac{6!}{2!2!2!} = 90$

Method: Three pairs: divide 720 by 2^3 .

Investigate further: Of these 90, how many are palindromes? (A palindrome is fixed by its first three digits, which must use each of 1, 2, 3 exactly once: $3! = 6$.) Symmetry constraints slash repeated-letter counts — try “how many read the same upside down” style variants.

8. Arrangements of LEVEL?

Answer: $\frac{5!}{2!2!} = 30$

Method: L $\times$ 2, E $\times$ 2, V: divide 120 by 4.

Investigate further: How many of the 30 are palindromes? (Middle letter forced to V; first two letters L, E in some order determine the rest: 2.) Notice palindrome counting always halves the alphabet workload: only half the word is free.

9. Seven books, trilogy in volume order (not necessarily adjacent)?

Answer: $\frac{7!}{3!} = 840$

Method: Forced-order trick: of the $3!$ orderings of the trilogy within any arrangement, exactly one is correct — so divide $7!$ by $3!$. No gluing: they need not be adjacent.

Investigate further: Compare the three trilogy conditions: adjacent-any-order ($3! \cdot 5! = 720$), adjacent-in-order ($5! = 120$), in-order-anywhere (840). Rank them by restrictiveness *before* computing next time. The divide-by-order trick is sheet 1’s “strictly increasing digits” insight, now industrialised.

10. Round-table seatings of n people = 720. Find n .

Answer: $n = 7$

Method: $(n - 1)! = 720 = 6!$, so $n = 7$.

Investigate further: Factorials are so sparse that “= 720” identifies $6!$ instantly — recall sheet 1’s ladder. Solve also $(n - 1)! \cdot 2 = 240$ (a glued-pair circle count): $n = 6$.

Section C Substitution & Structure

(≤ 90 s each)

1. Eight in a row: three friends consecutive, two others (non-friends) not adjacent?

Answer: $3! \cdot 6! - 3! \cdot 2! \cdot 5! = 4\,320 - 1\,440 = 2\,880$

Method: Layer the constraints (sheet's D2 pattern, arriving early). Glue the trio: $3! \cdot 6! = 4\,320$ arrangements. From these subtract those where the two enemies are also adjacent — glue them too: $3! \cdot 2! \cdot 5! = 1\,440$. Answer 2 880.

Investigate further: Generalise the single-constraint count: k specified people consecutive among n gives $k!(n - k + 1)!$ — check it against B1. Then redo this question with the gap method instead of the complement and confirm agreement.

2. Same eight, no refusals, friends consecutive *and* in height order?

Answer: $6! = 720$

Method: The block's internal order is now forced: glue with factor 1, then 6 objects in a row.

Investigate further: “Consecutive AND ordered” = glue with factor 1; “ordered but scattered” = divide by $3!$ (B9). Keep the two tricks distinct in your head — mixing them up is a classic exam slip.

3. Arrangements of DIVIDED without all three D's together?

Answer: $\frac{7!}{3!2!} - \frac{5!}{2!} = 420 - 60 = 360$

Method: Total arrangements: $\frac{7!}{3!2!} = 420$. All-D's-together: glue [DDD] (identical letters, factor 1) and arrange it with I, I, V, E: $\frac{5!}{2!} = 60$. Subtract.

Investigate further: Careful distinction: the complement of “all three together” still allows *two* D's adjacent. Count the arrangements where no two D's are adjacent at all (gap method on I, I, V, E: $\frac{4!}{2!} \cdot \binom{5}{3} = 120$) and check $120 < 360$ makes sense.

4. Four men, four women alternating around a round table?

Answer: $3! \cdot 4! = 144$

Method: Seat the men first: circular, so $(4 - 1)! = 6$. Their seating fixes 4 alternating gaps, into which the women go in $4!$ ways. (No extra factor 2: in a circle the two “patterns” are rotations of each other.)

Investigate further: Row version (B5) had the pattern factor 2; the circle ate it. Trace exactly where: rotating an MWMW... row by one seat turns it into WMWM... — rotations already identify the two patterns. Symmetries must be counted once, not twice.

5. Arrangements of CIRCLE with the two C's apart?

Answer: $\frac{6!}{2!} - 5! = 360 - 120 = 240$

Method: Total (with C repeated): 360. Together: glue CC (identical, no internal factor): $5! = 120$. Subtract.

Investigate further: Redo via gaps: arrange I, R, L, E ($4!$), choose 2 of the 5 gaps for the identical C's ($\binom{5}{2}$): $24 \cdot 10 = 240$. ✓ Complement and gaps agree — and for *identical* letters the gap count uses $\binom{}{}$, not a falling product. Spot why.

6. Five-digit even numbers over 40 000 using 1, 2, 3, 4, 5 once each?

Answer: 18

Method: Two interacting constraints (units even, leading digit 4 or 5) — case on the units digit, because it consumes a candidate leading digit or not. *Units 2:* leading digit from $\{4, 5\}$, rest free: $2 \cdot 3! = 12$. *Units 4:* leading digit must be 5, rest free: $3! = 6$. Total 18.

Investigate further: Why case on units rather than the leading digit? Try it the other way and watch the option-counts stop being uniform (sheet 1's C1 lesson). Extension: how many are divisible by 4? (Endings 12, 24, 32, 52; now impose the $> 40\,000$ condition on each case.)

7. Three maths, two physics, two chemistry books, subjects together?

Answer: $3! \cdot (3! \cdot 2! \cdot 2!) = 6 \cdot 24 = 144$

Method: Arrange the three subject blocks ($3!$), then the books inside each block ($3!, 2!, 2!$).

Investigate further: Blocks-of-blocks compose: shelves of subjects of books of chapters... each level multiplies independently. Now add the constraint “maths must be left of chemistry” — which factor changes, and to what? (The $3!$ becomes 3: divide by the $2!$ orderings of the M–C pair.)

8. Eight around a round table, two people refusing to be adjacent?

Answer: $7! - 2 \cdot 6! = 5\,040 - 1\,440 = 3\,600$

Method: All seatings ($7!$) minus glued-pair seatings ($2 \cdot 6!$).

Investigate further: Gap version: seat the 6 others round the table ($5!$), then put the two enemies into 2 of the 6 gaps: $5! \cdot 6 \cdot 5 = 3600$. ✓ Both routes again — build the habit of solving restricted seatings twice until they always agree first time.

Section D Challenge

(SMC difficulty)

1. Seven-digit numbers from 1–7: odds increasing, evens decreasing?

Answer: $\binom{7}{4} = 35$

Method: Choose which 4 of the 7 positions hold the odd digits: $\binom{7}{4} = 35$. Everything else is forced — the odds 1, 3, 5, 7 must fill their positions in increasing order (one way), and the evens 6, 4, 2 fill theirs in decreasing order (one way). A count that looks like it needs $7!$ cases collapses to a single choice.

Investigate further: Equivalently by double forced-order division: $\frac{7!}{4!3!} = 35$ — divide by the orderings of the odds AND of the evens. With only the odds constrained the count is $\frac{7!}{4!} = 210$; check both against the choose-positions view. This “order is forced, so only positions matter” insight is the SMC’s favourite disguise for $\binom{n}{k}$.

2. Ten children in a circle: Mia–Noah adjacent, Omar–Priya not?

Answer: $2 \cdot 8! - 4 \cdot 7! = 80\,640 - 20\,160 = 60\,480$

Method: Layer the constraints. Glue Mia–Noah: $2 \cdot (9 - 1)! = 80\,640$ circles. From these subtract the ones where Omar–Priya are also adjacent (glue both pairs): $2 \cdot 2 \cdot (8 - 1)! = 20\,160$. Answer: 60 480.

Investigate further: The pattern “force the easy constraint, complement the hard one” handles most two-constraint arrangement problems. Try three pairs: A–B adjacent, C–D adjacent, E–F *not*

adjacent, 10 people in a circle. (Same layering, one more glue.)

3. Arrangements of AABBC with no two identical letters adjacent?

Answer: 30

Method: Inclusion–exclusion on the “bad pair” events, computed cleanly by gluing. Total: $\frac{6!}{2!2!2!} = 90$. Glue AA: $\frac{5!}{2!2!} = 30$, same for BB, CC. Glue two pairs: $\frac{4!}{2!} = 12$ each (3 ways). Glue all three: $3! = 6$. Count with no glued pair = $90 - 3 \cdot 30 + 3 \cdot 12 - 6 = 30$.

Investigate further: This is inclusion–exclusion arriving a sheet early — sheet 5 makes it systematic. For now, verify the answer differently: by symmetry all $3!$ letter-labelings are equivalent, so just count patterns like ABCABC, ABCACB, ... and multiply. How many *patterns* are there? ($30/3! = 5$.)

4. Eight in a row, exactly two of the three sisters adjacent?

Answer: $8! - 14\,400 - 4\,320 = 21\,600$

Method: Three disjoint outcomes partition all $8! = 40\,320$ arrangements: *no* adjacent sister-pair, *exactly one* adjacent sister-pair, and *two* adjacent sister-pairs (which is precisely “all three consecutive”). *None adjacent* (gap method): $5! \cdot 6 \cdot 5 \cdot 4 = 14\,400$. *All three consecutive*: $3! \cdot 6! = 4\,320$. “Exactly two sisters adjacent” is the middle category, so it equals $40\,320 - 14\,400 - 4\,320 = 21\,600$.

Investigate further: Check the borderline case that makes this work: in a block of three sisters, the outer two are NOT adjacent to each other — adjacency is about neighbouring seats, not membership of a block. Redo the problem for “at least two adjacent” ($40\,320 - 14\,400 = 25\,920$) and confirm the three-way split is consistent.

5. Eight knights round a table: three feuders pairwise non-adjacent, King and Queen together?

Answer: $3! \cdot 2 \cdot (4 \cdot 3 \cdot 2) = 288$

Method: Layer: glue first, then gaps. Glue the King–Queen pair (2 internal orders); the peaceful side is now 4 units (the KQ block plus 3 other knights). Seat those 4 units round the table: $(4 - 1)! = 6$ circular orders, creating exactly 4 gaps. Place the three feuding knights into three different gaps (one each — sharing a gap would make two feuders adjacent): $4 \cdot 3 \cdot 2 = 24$. Total $6 \cdot 2 \cdot 24 = 288$.

Investigate further: Check the order of operations matters: try placing the feuders first and gluing second, and watch the gap count become ill-defined — constraints that *create* structure (glue) must precede constraints that *consume* it (gaps). Then generalise the no-royals version: n knights, k feuding, count = $(n - k - 1)! \cdot \frac{(n - k)!}{(n - 2k)!}$; test it at $n = 8, k = 3$ (1440) against a hand count of a small case like $n = 5, k = 2$.

Top 5 Patterns Today

- Glue blocks.** “Must be together” $\Rightarrow$ treat as one object, multiply by internal orders (which is 1 if the glued items are identical or their order is forced).
- Count gaps.** “No two adjacent” $\Rightarrow$ seat the unrestricted objects, then place the restricted ones into distinct gaps. Rows of m give $m + 1$ gaps; circles of m give m gaps.
- Divide by forced order.** If k objects must appear in one specified relative order (not necessarily adjacent), count freely and divide by $k!$.
- Fix a seat in circles.** Rotational symmetry costs one factor of n : circular $n = (n - 1)!$. Ask

separately whether reflections are also “the same”.

- 5. Layer constraints.** Force the “must” constraints by gluing first, then handle the “must not” by complement or gaps inside that restricted world.

Common Traps to Avoid

- 1. Multiplying by internal orders of identical letters.** Gluing the two E’s of SPEED is a factor 1, not 2. Only *distinguishable* contents of a block get a factorial.
- 2. Row gaps vs circle gaps.** Five people create 6 gaps in a row but only 5 in a circle. Using the row count at a round table silently overcounts.
- 3. Double-counting the alternating patterns in a circle.** MWMW... and WMWM... are rotations of each other at a round table — the row-version factor 2 must be dropped (C4).
- 4. Confusing “in order” with “adjacent”.** Volume order scattered along the shelf divides by 3!; volumes boxed together glues a block. Different tricks, different answers (B9 vs C1/C2).
- 5. Treating “exactly one adjacent pair” as total minus none-adjacent.** That difference counts *at least one* pair. The all-together case must be carved out separately (D4).

“Glue what must be together, seat what must be apart last — the gaps will tell you how much room rebellion really has.”

other prep to look at today:
Daily Integral for Calculus and Logic